Unit 13: The Chain Rule

1. The idea, in plain words

So far, every function you've differentiated has had x sitting "bare" inside — like x^4, or $\sin x$. But what happens when there's a whole expression tucked inside another function, like $(x^2+1)^4$ or $\sin(3x)$? You can't just apply the power rule or the trig formulas directly to the outside and call it done — you have to account for the fact that the "inside" is doing its own thing too.

Picture it like a set of Russian nesting dolls. $(x^2+1)^4$ has an outer layer (something to the 4th power) wrapped around an inner layer (x^2+1). To differentiate the whole thing, you differentiate the outer layer first — pretending the inside is just a single blob — and then, because that inside blob isn't actually just "x," you have to multiply by the inside's own derivative too. That extra multiplication is the entire chain rule in one sentence.

Another way to picture it: imagine you're tracking how fast your friend's friend's friend is influencing you — a chain of connections. If A affects B, and B affects C, and C affects you, then the total effect of A on you is the product of each link's individual effect: (effect of A on B) × (effect of B on C) × (effect of C on you). The chain rule works the exact same way — you multiply together the "rate of change" at every layer of the nesting.

The most common version you'll use constantly — the Power Chain Rule: whenever you see anything raised to a power — not just plain x, but a whole expression u — the rule is: bring the exponent down, drop the exponent by one, and then multiply by the derivative of whatever's inside (u'). That last step, multiplying by u', is the one people forget the most — and it's the entire point of this unit.

2. Toolbox

The Chain Rule (general form): if $y = f(g(x))$ — an "outer" function f wrapped around an "inner" function g — then

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

In plain words: derivative of the outside (leaving the inside untouched) times derivative of the inside.

Power Chain Rule (the version you'll use most often — here u stands for any expression involving x):

$$\frac{d}{dx}[u^n] = n\,u^{n-1} \cdot u'$$

Trig Chain Rule versions (again, u is any inner expression, not just plain x):

$$\frac{d}{dx}[\sin u] = \cos u \cdot u' \qquad \frac{d}{dx}[\cos u] = -\sin u \cdot u'$$

$$\frac{d}{dx}[\tan u] = \sec^2 u \cdot u' \qquad \frac{d}{dx}[\cot u] = -\csc^2 u \cdot u'$$

$$\frac{d}{dx}[\sec u] = \sec u \tan u \cdot u' \qquad \frac{d}{dx}[\csc u] = -\csc u \cot u \cdot u'$$

How to identify the layers: ask yourself, "what's the very last operation being done to x?" That outermost operation is your "outer function." Whatever's left inside it is your "inner function," u.

Combining with earlier rules: the chain rule doesn't replace the product or quotient rule — it works alongside them. If a problem has both a product and a composition, you'll need to use both rules together, one at each spot where they're needed.

3. Common mistakes

- Forgetting to multiply by the derivative of the inside — by far the most common mistake in this entire unit. Differentiating $(x^2+1)^4$ as just $4(x^2+1)^3$ and stopping there is incomplete — you must still multiply by the derivative of x^2+1, which is $2x$.
- Applying the chain rule when it isn't needed. If the inside is just plain x (like x^4 by itself), there's no extra multiplication needed — the "derivative of the inside" is just 1, which doesn't change anything. The chain rule matters when the inside is something other than plain x.
- Mixing up which function is "outer" and which is "inner." Always ask what the very last operation applied to x is — that's the outer layer.
- Forgetting to also use the product or quotient rule when a problem needs both. A function like $x^2\sin^4 x$ needs the product rule (for the two factors x^2 and $\sin^4 x$) and the chain rule (to differentiate $\sin^4 x$ itself).
- Sign errors when the inner function involves subtraction or a trig function. Take extra care finding u' before you multiply it in.

4. Problem Set

Warm-up

1. $y=(x+3)^4$
2. $y=(2x-1)^5$
3. $y=(3x+2)^3$
4. $y=(x^2+1)^3$
5. $y=(5-x)^4$
6. $y=(4x+7)^2$
7. $y=(x^2-2x)^2$

Standard

8. $y=\sin(3x)$
9. $y=\cos(x^2)$
10. $y=(2x^2-3)^{-1}$
11. $y=\sqrt{x^2+4}$
12. $y=\tan(2x)$
13. $y=(3x-1)^{1/3}$
14. $y=\sin^2 x$ (that is, $(\sin x)^2$)

Challenge

15. $y=\left(\dfrac{x^2}{2}+x-\dfrac{1}{x}\right)^4$
16. $y=\sin^4 x$

17. $y = x^2 \sin^4 x + x \cos^{-2} x$
18. $y = (x^2+1)^3(x-1)^2$
19. $y = \sqrt{\sin x}$
20. $y = (1+\cos x)^3$

Applied

21. A balloon's radius (in cm) grows according to $r(t) = 2t$, and its volume is $V(r) = \frac{4}{3}\pi r^3$. Using the chain rule, find $\frac{dV}{dt}$, then evaluate it at $t = 1$.
22. A wave's height (in cm) is modeled by $h(t) = 3\sin(2t)$. Find $h'(t)$ using the chain rule, then evaluate $h'(0)$ and interpret what it means.
23. A company's production level (in units) is $x(t) = 10t$, and its cost (in dollars) is $C(x) = \sqrt{x^2+50}$. Using the chain rule, find $\frac{dC}{dt}$, then evaluate it at $t = 2$.
24. A greenhouse's temperature (in °F) is modeled by $T(t) = (t+2)^3 - 5$. Find $T'(t)$ using the chain rule, then evaluate $T'(1)$.